

10 (**long**)

The symmetric matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \frac{1}{4} \begin{bmatrix} 3 & -1 & 0 \\ -1 & 3 & 0 \\ 0 & 0 & 4a \end{bmatrix}$$

where a is a constant.

- (a) Determine the eigenvalues and eigenvectors of $\mathbf{A}$. [10]
- (b) By expressing $\mathbf{A}$ as the product of two orthogonal matrices and a diagonal matrix, determine an expression for $\mathbf{A}^n$ for any positive integer n . [5]
- (c) An approximate expression for $\mathbf{A}^n$ is required for large n and positive a . By examining the dominant eigenvalues, determine the different ranges of a for which $\mathbf{A}^n$ can be approximated. Give the different approximations for $\mathbf{A}^n$ with justification. [15]